

WASTE WATER TREATMENT METHODS

WASTEWATER TREATMENT

- The purpose of waste water treatment is to remove contaminant from water so that the treated water can meet the acceptable quality standard.
- Treatment process are broadly classified as
 - **Physical treatments**
 - **Chemical treatments**
 - **Biological treatment**

WASTE WATER TREATMENT PROCESS

1. Preliminary Treatment
2. Primary Treatment
3. Secondary or Biological Treatment
4. Tertiary or Advanced Treatment

1. PRELIMINARY TREATMENT OF SEWAGE

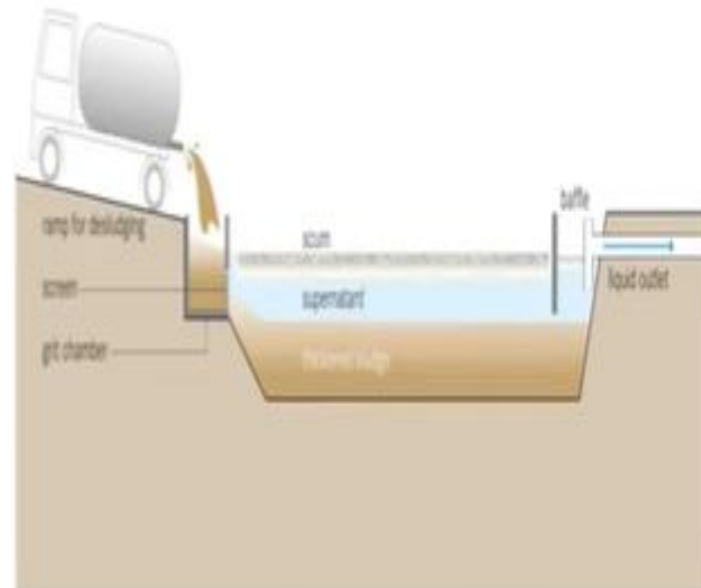
- The main objective of preliminary treatment is **to remove solids** from waste water.
- Some techniques applied for preliminary treatment purpose are:
 - (i). Screening, (ii). Grit removal, (iii). Skimming.**

i. Screening:

- Screening is used to remove solid waste like plastics, cloths, dead animals from waste water. In screening, waste water is passed through **screener**. The screeners are classified as **coarse, medium and fine**, depending on the size of the openings.

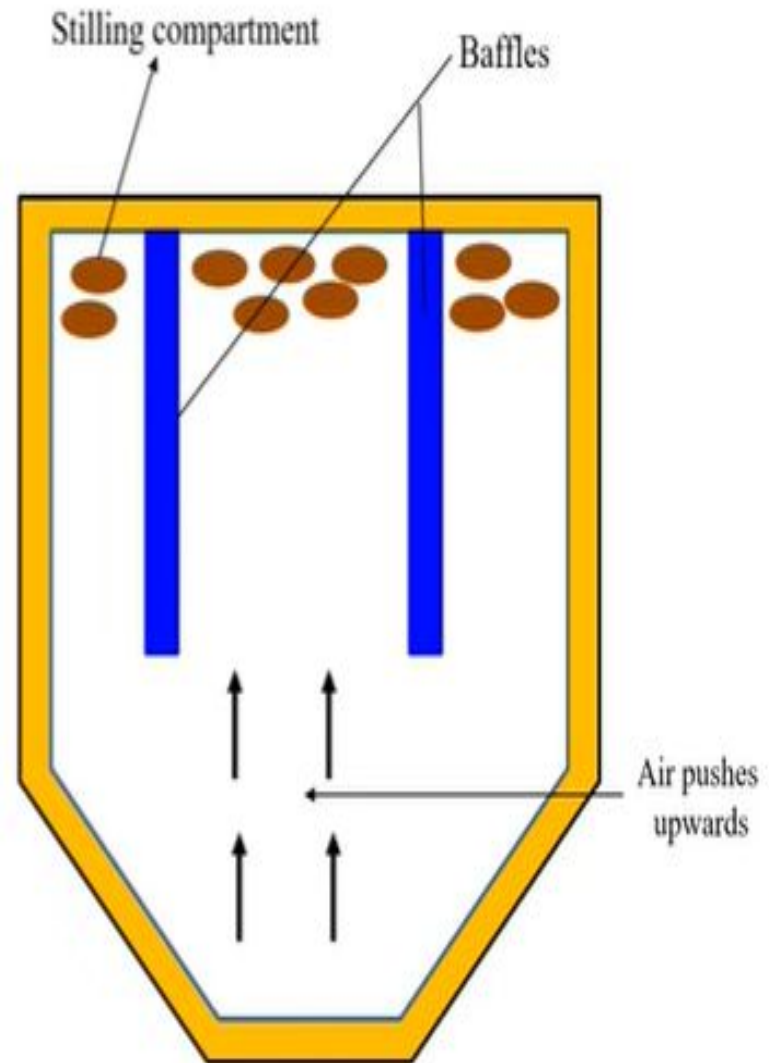
ii. Grit Removal

- Grits are small, non-biodegradable particles which are heavier than suspended organic matters.
- The heavy inorganic materials like sand, ash and others can be removed by using **grit chambers**.



iii. Skimming

- Skimming is the **process of removal of fatty and oily material** from sewage. In this method, sewage water is placed in **skimming tank**.



Skimming tank.

2. PRIMARY TREATMENT OF SEWAGE

- Primary treatment remove fine suspended organic solids that cannot be removed in the preliminary treatment.
- Some techniques applied for primary treatment of sewage are **(i). Sedimentation, (ii). Mechanical flocculation, (iii). Chemical flocculation.**

i. Sedimentation:

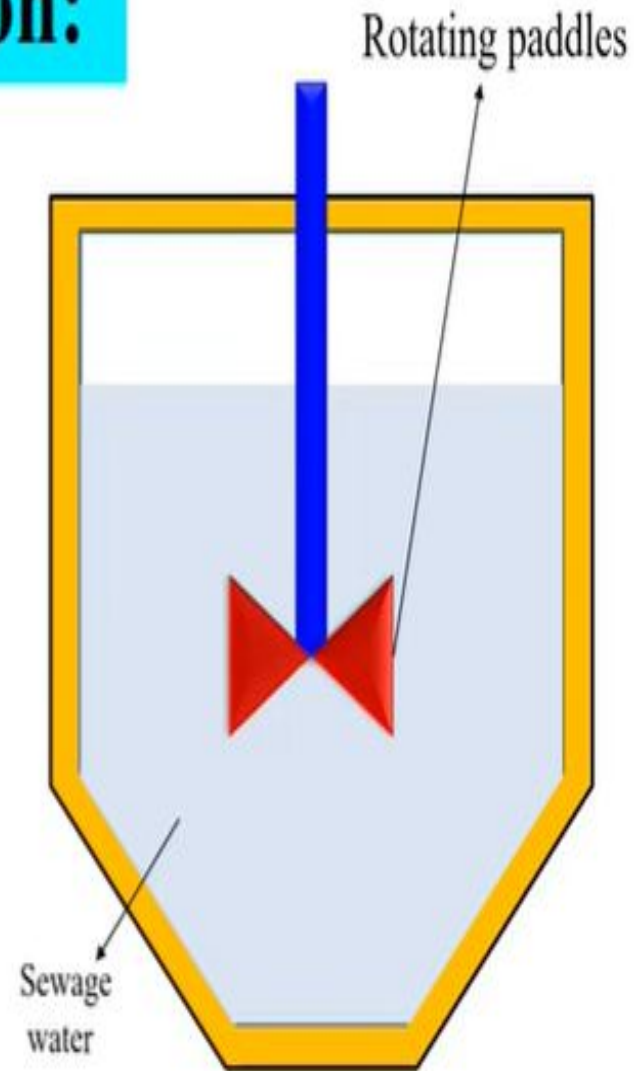
The solid particle of the sewage tend to settle down due to gravity. Sedimentation is done in **sedimentation tank.**



Sedimentation tank

ii. Mechanical flocculation:

In this method sewage is placed in a **flocculation tank**, then sewage is rotated at an speed of 0.43m/sec with the help of rotating paddles.



Flocculation tank.

iii. Chemical flocculation:

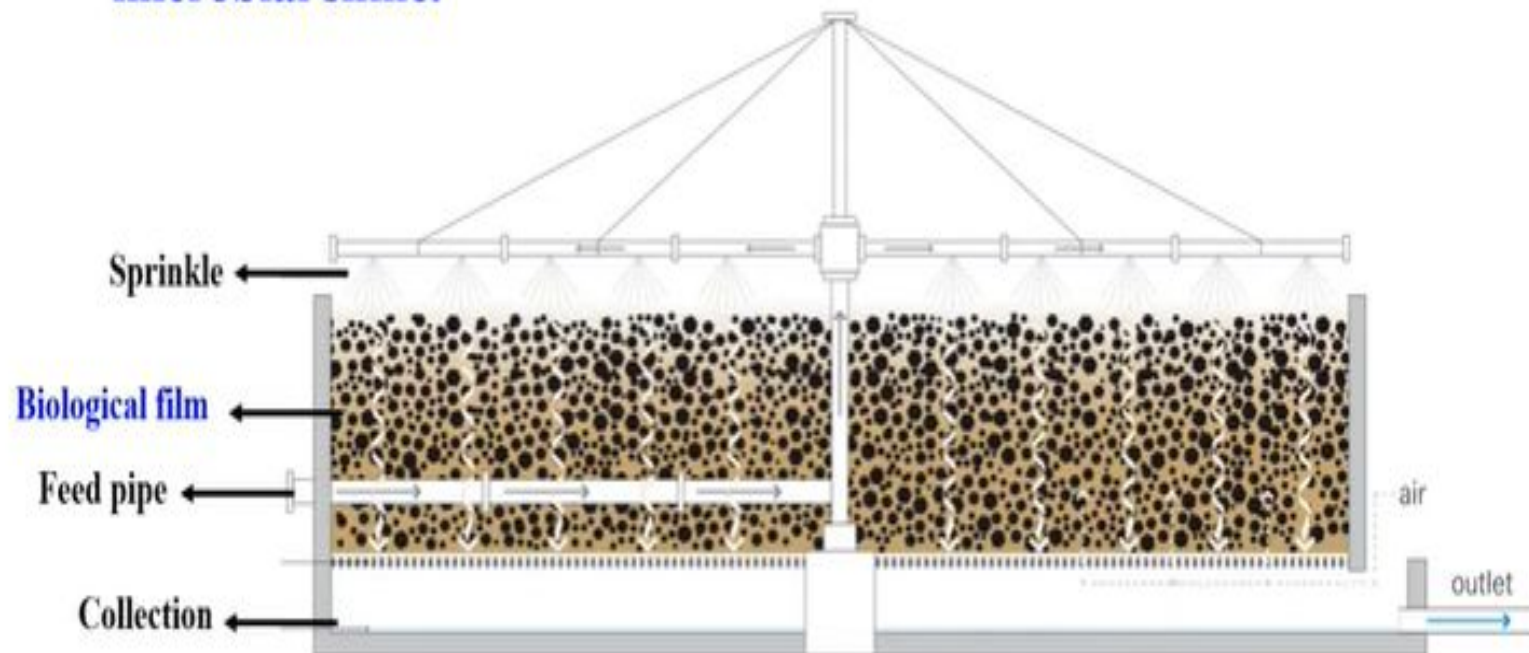
- In this method, sewage is placed in coagulation tank and then some precipitating agents such as **alum** is added. **Alum** forms precipitate of Aluminium hydroxide $[\text{Al}(\text{OH})_3]$.

3. SECONDARY OR BIOLOGICAL TREATMENT OF SEWAGE

- Biological treatment involves the use of microorganisms (bacteria, algae, fungi, protozoa, nematodes) that decompose the unstable organic matter to stable inorganic forms.
- Some techniques applied for primary treatment of sewage are (i). **Trickling filter**, (ii). **Activated sludge system**, (iii). **Oxidation ditch**.

i. Trickling filter:

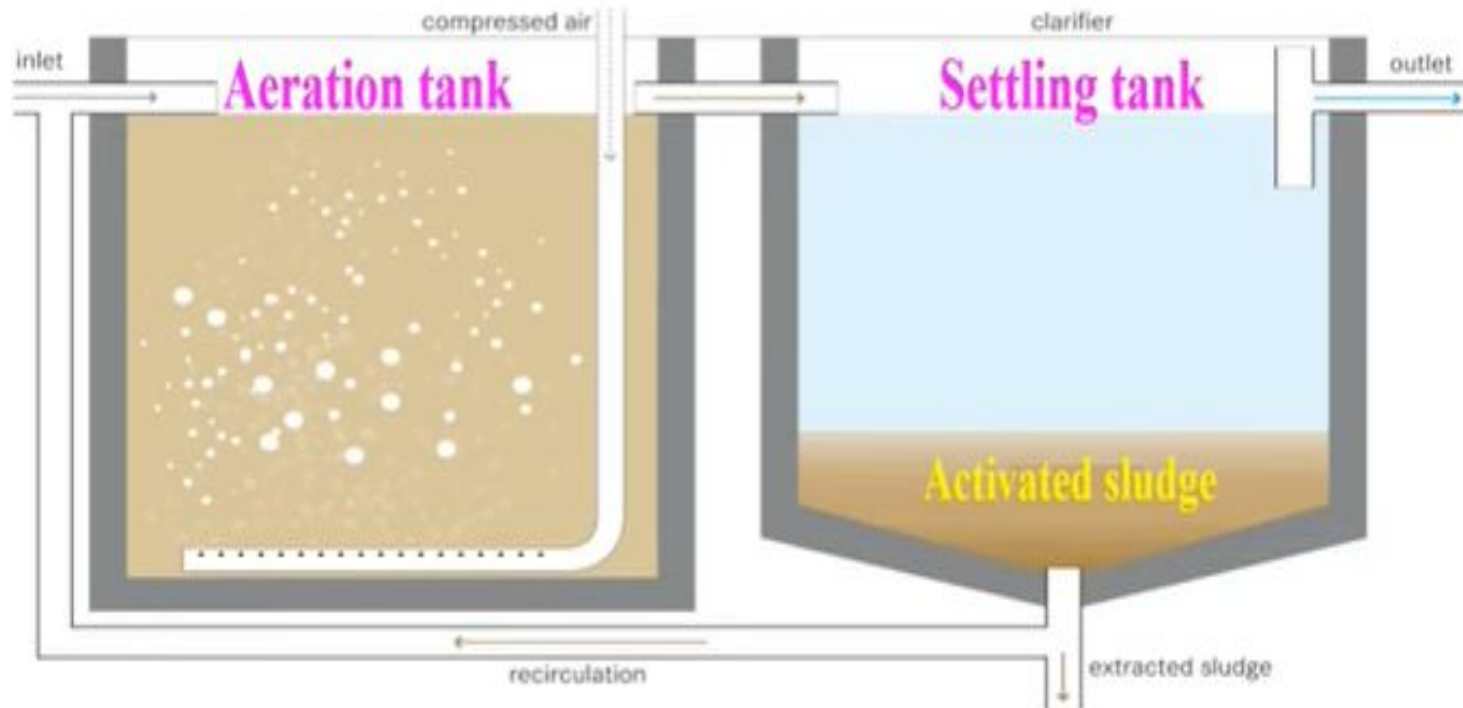
- It has a **bed of coarse, hard and porous material** over which sewage water is sprayed. The biomass *i.e.* microorganisms attached to the media surface grows and forms a layer, referred to as **biological film or microbial slime**.



- As the sewage water trickles through the biofilm, the organic matter gets oxidized to CO_2 and NO_2 by the microbial metabolism. This oxidation is carried out by the aerobic organisms that are present on the upper portion of the biological film.
- The biological film is rich in the bacteria *Pseudomonas*, *Flavobacterium*, *Alcaligenes*, and algae *Chlorella*, *Ulothrix*, and *Stigeoclonium*.

ii. Activated sludge system:

- Activated sludge system, consists of aeration tank, settling tank and sludge return system. The activated sludge contains large number of microorganisms and serves as inoculum of microorganisms.



iii. Oxidation ditch:

- Oxidation ditch consists of circular canal with inlet and outlet.
- In this method, sewage from primary treatment plant is placed in oxidation ditch and then it is agitated with the help of mechanical rotator and then left for a period of about 12-24 hours.



Inlet of oxidation pond

3. TERTIARY TREATMENT OF SEWAGE

- The main objectives of tertiary treatment process are
 - i. **Removal of suspended solids,**
 - ii. **Removal of dissolved solids like salts,**
 - iii. **Removal of nitrate and phosphate,**
 - iv. **Killing of microorganisms.**

iii. Removal of nitrate and phosphate:

- Eutrophication is the process by which a body of water becomes overly enriched with minerals and nutrients which induce excessive growth of algae.
- At first sewage is placed in a tank containing **nitrifying bacteria** like *Nitrosomonas*, *Nitrosococcus*, *Nitrobacter* etc. These bacteria convert ammonium salt and nitrite into nitrate. Then the sewage is placed into second tank containing **denitrifying bacteria** like *Thiobacillus denitrificans*, *Micrococcus denitrificans* etc. These bacteria convert nitrate into nitrogen gas that leaves the sewage. Phosphate is also removed by bacteria (*Pseudomonas* and *Bacillus* etc.) by **microbial absorption process**.

iv. Killing of microorganisms:

- Finally microorganisms in sewage are killed by **disinfection**.

A. Chemical agents:

- Chlorine is most commonly used.

B. Physical agents:

- Heat and light (ultra-violet rays).